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- Multiplying each element of a row by the same non-zero scalar
- Change a row by adding it to a multiple of another row
- Note the similarities of these operations to the operations we learn to use when solving systems of equations without matrix algebra.

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- The leading coefficient (the first non-zero element from the left) of a non-zero row is always strictly to the right of the leading coefficient of the row above it.
- All entries in a column below a leading entry are zeros (this condition is implied by the other two).

## Row Echelon Form: Examples

Which of the following matrices are in Row Echelon Form?

$$A = \begin{pmatrix} 4 & 2 \\ 0 & 7 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 & 10 \\ 0 & 8 & -3 \\ 0 & 0 & 6 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 6 & 7 \\ 0 & 0 & 8 \end{pmatrix}, D = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}$$

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Note that  $D$  would be in row echelon form if we switched rows.

## Row Echelon Form with Elementary Matrix Operations

To get a matrix in row echelon form we can perform elementary matrix operations on the rows. For example,

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$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 8 & 6 \\ 1 & 1 & 1 \end{pmatrix} \quad A_1 = \begin{pmatrix} 0 & 1 & 2 \\ 4 & 8 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

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$$A_1 = \begin{pmatrix} 0 & 1 & 2 \\ 4 & 8 & 6 \\ 1 & 1 & 1 \end{pmatrix} \quad A_2 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 4 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

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- Subtract 4 times the third row from the second row

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$$A_2 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 4 & 2 \\ 1 & 1 & 1 \end{pmatrix} \quad A_3 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & -6 \\ 1 & 1 & 1 \end{pmatrix}$$

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- Subtract the third row from the first row
- Subtract 4 times the third row from the second row
- Subtract 4 times the first row from the second row
- Rearrange the rows to get row echelon form

# Solving Linear System of Equations

We can solve a system of equations by writing the matrix

$$\left( \begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{nn} & b_m \end{array} \right)$$

This is called the *augmented matrix*.

## Solving Linear System of Equations: Example

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 8 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

and say the vector  $\mathbf{b} = (1, 1, 1)'$ . So we have the system

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 8 & 6 \\ 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

or  $\mathbf{b} = A\mathbf{x}$

The the augmented matrix is

$$\left( \begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 8 & 6 & 1 \\ 1 & 1 & 1 & 1 \end{array} \right)$$

## Solving Linear System of Equations: Example continued

Performing the same matrix operations as before, we have

$$\begin{aligned} & \left( \begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 8 & 6 & 1 \\ 1 & 1 & 1 & 1 \end{array} \right) \Rightarrow \left( \begin{array}{ccc|c} 0 & 1 & 2 & 0 \\ 4 & 8 & 6 & 1 \\ 1 & 1 & 1 & 1 \end{array} \right) \\ & \Rightarrow \left( \begin{array}{ccc|c} 0 & 1 & 2 & 1 \\ 0 & 4 & 2 & -3 \\ 1 & 1 & 1 & 1 \end{array} \right) \Rightarrow \left( \begin{array}{ccc|c} 0 & 1 & 2 & 0 \\ 0 & 0 & -6 & -3 \\ 1 & 1 & 1 & 1 \end{array} \right) \\ & \Rightarrow \left( \begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & -6 & -3 \end{array} \right) \end{aligned}$$

## Solving Linear System of Equations: Example

Continue until LHS is the identity matrix

$$\begin{aligned} \left( \begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & -6 & -3 \end{array} \right) &\Rightarrow \left( \begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right) \Rightarrow \left( \begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right) \\ &\Rightarrow \left( \begin{array}{ccc|c} 1 & 1 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right) \Rightarrow \left( \begin{array}{ccc|c} 1 & 0 & 0 & \frac{3}{2} \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right) \\ \left( \begin{array}{c} \frac{3}{2} \\ -1 \\ \frac{1}{2} \end{array} \right) &= \left( \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \cdot \left( \begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right) \Rightarrow \mathbf{x} = \left( \frac{3}{2}, -1, \frac{1}{2} \right) \end{aligned}$$

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If we form the augmented matrix of  $A$ , reduce the left hand side to its reduced row echelon form through elementary row operations, then the remaining vector on the right hand side will be the solution to the system, although it may not be a unique solution.

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- **Cramer's Rule** the unique solution  $\mathbf{x} = (x_1, \dots, x_n)$  of the  $n \times n$  system  $A\mathbf{x} = \mathbf{b}$  is

$$x_i = \frac{\det B_i}{\det A}, \text{ for } i = 1, \dots, n,$$

where  $B_i$  is the matrix  $A$  with the right-hand side  $\mathbf{b}$  replacing the  $i$ th column of  $A$

## Cramer's Rule: Example

Consider the system  $\mathbf{b} = A\mathbf{x}$ :

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$B_2$ ?  $B_2 = \begin{pmatrix} 1 & 1 & 3 \\ 4 & 1 & 6 \\ 1 & 1 & 1 \end{pmatrix}$ . What is  $x_2$ ?  $x_2 = \frac{-6}{6} = -1$

# Linear Independence

**Linear Independence** The vectors  $v_1, \dots, v_m$  are *linearly dependent* if there exist scalars  $q_1, \dots, q_m$ , not all zero such that

$$\sum_{i=1}^m q_i v_i = 0$$

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This is equivalent to the statement that none of the vectors can be written as a linear combination of the other vectors. Finally, note that we can use this definition for the columns or rows of a matrix.

## Spanning Vectors

A set of vectors **spans** a vector space (a set of vectors obeying vector addition and scalar multiplication) if the vector space can be written as a linear combination of the vectors from the (spanning) set of vectors.

- The set of vectors  $(1, 0)$  and  $(0, 1)$  span 2-D space. Any vector on a Euclidean plane can be written as a linear combination of  $(1, 0)$  and  $(0, 1)$ .

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- $n$  linearly independent vectors must span the  $n$  Dimension vector space.

## Linear Independence: Example

Are these vectors linearly dependent

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These vectors are linearly independent. The sum of these vectors only equals zero when  $q_1 = q_2 = q_3 = 0$ .

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- A matrix has full rank if:
  - $rk(A) = \min\{\text{number of rows of } A, \text{number of columns of } A\}$

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is linearly independent because its row echelon form is

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Note that if there is a row of all zeros in reduced row echelon form we know either that the row was originally all zeros or it was a linear combination of the other rows.

# Characterizing the Number of Solutions to a Linear System

The locus of all points in  $(x_1, x_2)$ , which satisfy the linear equation  $a_{11}x_1 + a_{12}x_2 = b_1$  is a straight line in the plane. Therefore, the solution  $(x_1, x_2)$  of the two linear equations in two unknowns

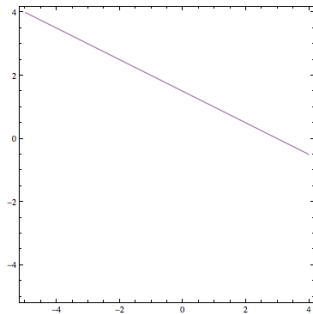
$$a_{11}x_1 + a_{12}x_2 = b_1$$

$$a_{21}x_1 + a_{22}x_2 = b_2$$

is a point that lies on both lines of these equations. Solving these systems is equivalent to finding where the lines given by the equations cross. If the lines are parallel, then there is no solution to the system. If the lines lie on top of each other, there are infinitely many solutions.

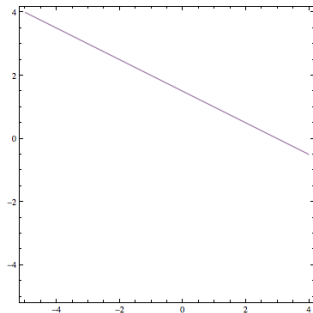
# Solutions to a Linear System: Example 1

$$\begin{aligned}x_1 + 2x_2 &= 3 \\ 2x_1 + 4x_2 &= 6\end{aligned}$$



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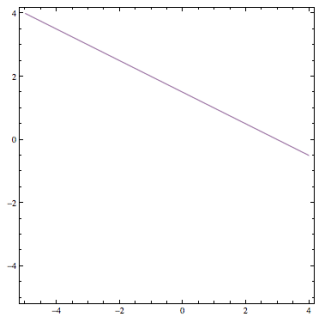
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What is particular about this system of equations?

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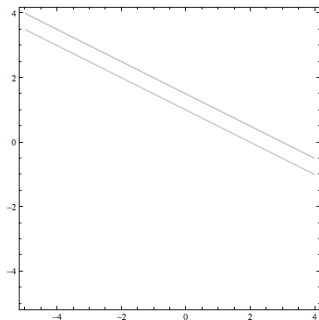
What is particular about this system of equations? If expressed in matrix terms the matrix  $(A|\mathbf{b})$  has a rank of only 1, its rows are linearly dependent.



## Characterizing the Number of Solutions to a Linear System: Example 2

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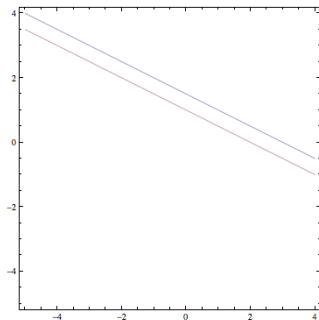
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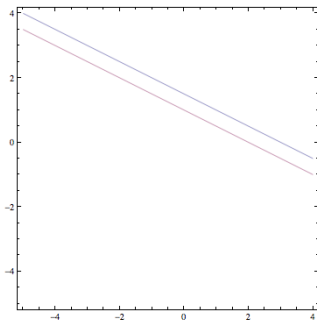


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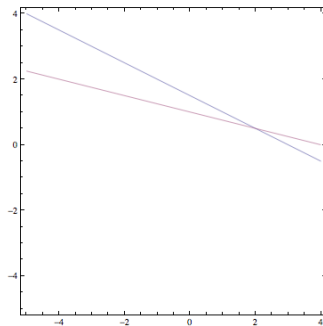


What is particular about this system of equations? If expressed in matrix terms, the matrix ( $A$ ) has a rank of only 1, its rows are linearly dependent, which means the lines will be parallel.

## Solutions to a Linear System: Example 3

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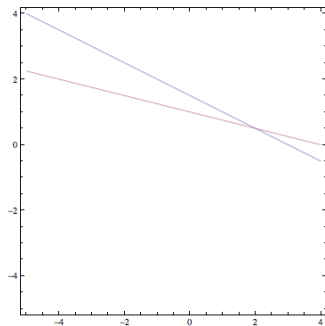
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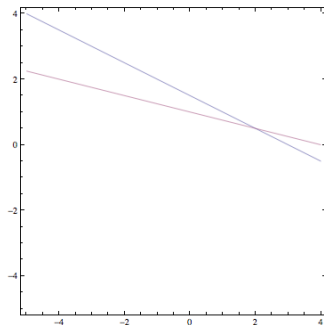


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## Solutions to a Linear System: Example 3

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What is particular about this system of equations? If expressed in matrix terms, the matrix ( $A$ ) has a rank of 2, its rows are linearly independent.

## Solutions to a Linear System

Consider a system of  $m$  equations with  $n$  unknowns, characterized by  $\mathbf{b} = A\mathbf{x}$ . (This means that  $A$  will be an  $m \times n$  matrix.) If  $m < n$ , then:

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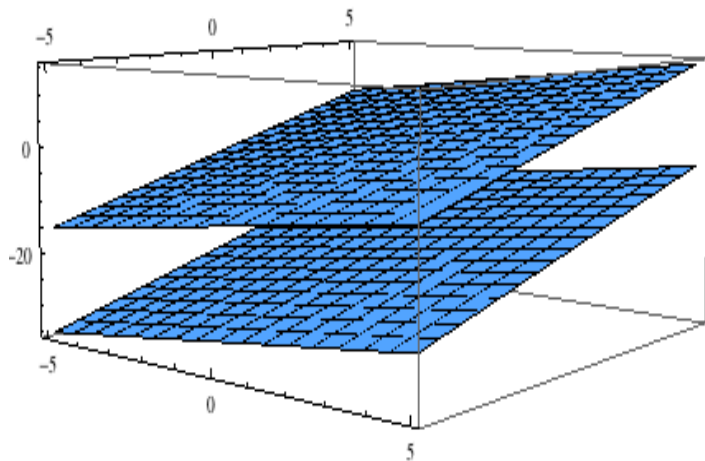
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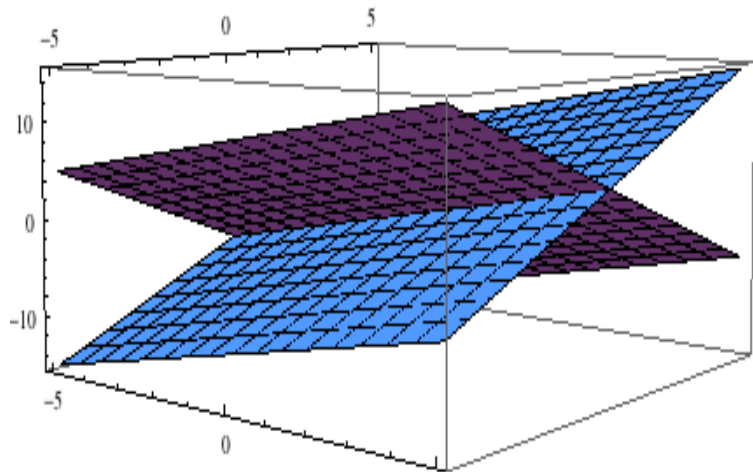
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When a system has more unknowns than equations, it can never have a unique solution. For example, two planes in  $\mathbb{R}^3$  cannot intersect at only one point. They either are parallel and have no solution or cross and have infinitely many solutions. If  $A$  has maximal rank of  $m$  then we know that the system has infinitely many solutions. For example in  $\mathbb{R}^3$  the planes would cross yielding infinitely many solutions. We also know that the planes aren't identical but there are still an infinite amount of solutions.

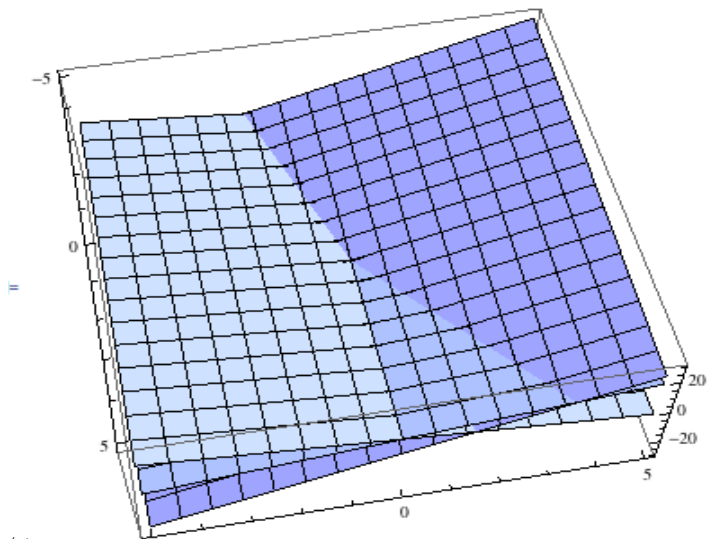
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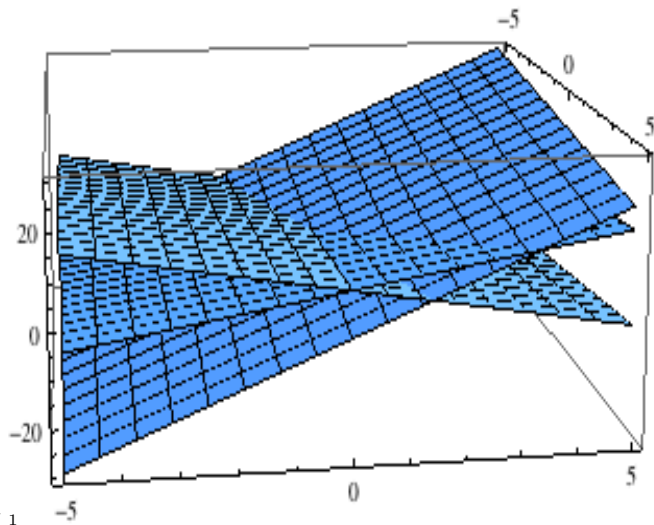
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A system with more equations than unknowns may have zero, one or infinitely many solutions. For example, three lines in  $\mathbb{R}^2$  can all not intersect (no solution), can intersect at one point (one solution) or can all be identical (infinitely many solutions). If we know that  $A$  has maximal rank of  $n$  then the system cannot have infinitely many solutions. We know this because all of the lines cannot be parallel or identical since they aren't linear complements. Still the system might have zero solutions if the lines do not cross at the same point.

# Solutions to Linear System

Finally consider  $m = n$ , then

- For any given  $\mathbf{b}$ ,  $\mathbf{b} = A\mathbf{x}$  has zero, one or infinitely many solutions.
- If  $rkA = m = n$ ,  $\mathbf{b} = A\mathbf{x}$  has exactly one solution for every  $\mathbf{b}$

A system with an equal number of equations and unknowns can have zero, one, or infinitely many solutions. For example, two lines in  $\mathbb{R}^2$  can be parallel, intersect at one point or be identical. (Same for three planes in  $\mathbb{R}^3$ .) If we know that  $A$  has maximal rank then we know the system has a unique solution. In our example, we know that the lines are not parallel and must cross at a particular point.

## Relationship between the Determinant and Linear System

An  $n \times n$  square matrix  $A$  has full rank  $n$  if and only if  $\det A \neq 0$ . This means that the  $n \times n$  system  $\mathbf{b} = A\mathbf{x}$  has exactly one solution for each  $\mathbf{b}$  if and only if  $\det A \neq 0$ . This means that if a matrix  $A$  is nonsingular it is equivalent to saying that  $rkA = n$ , which is equivalent to saying  $\mathbf{b} = A\mathbf{x}$  has exactly one solution for each  $\mathbf{b}$ . Finally note that when  $\det A = 0$  Cramer's rule fails.

# Inverse Matrices and Linear Systems

Suppose we are given a  $n \times n$  matrix  $A$ . Suppose that the inverse  $A^{-1}$  exists. Then we can multiply both sides of  $\mathbf{b} = A\mathbf{x}$  by  $A^{-1}$ :

$$A^{-1}\mathbf{b} = A^{-1}A\mathbf{x} = I\mathbf{x} = \mathbf{x}$$

Thus  $\mathbf{x} = A^{-1}\mathbf{b}$  has to be the unique solution to  $\mathbf{b} = A\mathbf{x}$ .

Conversely, it can also be shown that if there is a unique solution to  $\mathbf{b} = A\mathbf{x}$ , then the inverse  $A^{-1}$  has to exist. (For the proof see S and B page 167). Therefore, a square matrix is invertible if and only if it is nonsingular, etc...

The following statements are all equivalent for a square matrix  $A$ :

- $A$  is non-singular
- All the columns and rows in  $A$  are linearly independent.
- $A$  has full rank
- Exactly one solution  $\mathbf{X}^*$  exists for each vector  $\mathbf{Y}^*$
- $A$  is invertible
- $\det(A) \neq 0$
- The row-echelon form of the matrix is upper triangular.
- The reduced row echelon form is the identity matrix

# Calculating the Inverse Matrix by Elementary Row Operation

To calculate the inverse matrix, form the augmented matrix:

$$\left( \begin{array}{cccc|cccc} a_{11} & a_{12} & \dots & a_{1n} & 1 & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & a_{2n} & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & 0 & 0 & \dots & 1 \end{array} \right)$$

Reduce the left hand side to the reduced row echelon form, and what remains on the right hand side will be the inverse matrix of  $A$ . In other words, using elementary row operations you can transform the matrix  $(A|I)$  to  $(I|A^{-1})$ .

# Quadratic Form

Consider the function  $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ , where

$F = a_{11}x_1^2 + a_{12}x_1x_2 + a_{22}x_2^2$ . We call this a *quadratic form* in  $\mathbb{R}^2$ .

Notice that this can be expressed in matrix form as

$$F(\mathbf{x}) = (x_1 \ x_2) \begin{pmatrix} a_{11} & \frac{1}{2}a_{12} \\ \frac{1}{2}a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \mathbf{x}' A \mathbf{x}$$

where  $\mathbf{x}$  is a column and  $A$  is unique and symmetric.



## Quadratic Form in $\mathbb{R}^n$

The quadratic form in  $\mathbb{R}^n$  is

$$F(\mathbf{x}) = \sum_{i,j=1}^n a_{ij}x_i x_j$$

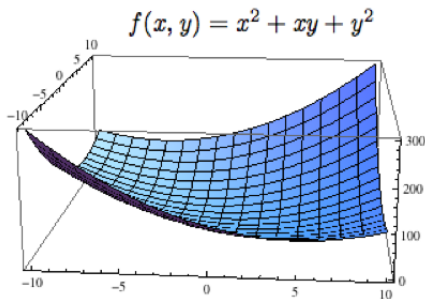
This can also be expressed in matrix form

$$F(\mathbf{x}) = (x_1 x_2 \dots x_n) \begin{pmatrix} a_{11} & \frac{1}{2}a_{12} & \dots & \frac{1}{2}a_{1n} \\ \frac{1}{2}a_{12} & a_{22} & \dots & \frac{1}{2}a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{2}a_{1n} & \frac{1}{2}a_{2n} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \mathbf{x}' A \mathbf{x}$$

Where  $\mathbf{x} = (x_1, \dots, x_n)$  and  $A$  is symmetric.

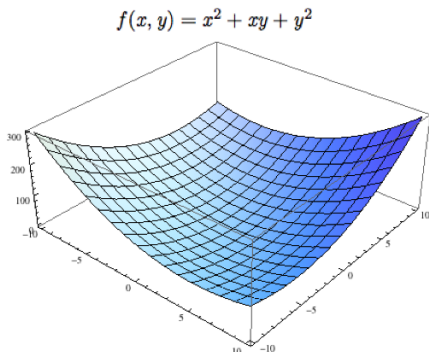
## Definiteness

A quadratic form has a critical point at  $\mathbf{x}$ , where it takes on the value 0. We can thus, classify quadratic forms by whether  $\mathbf{x} = 0$  is a maximum, minimum, or neither. The properties of definiteness provide this information. To first get some intuition consider the following simple quadratics.



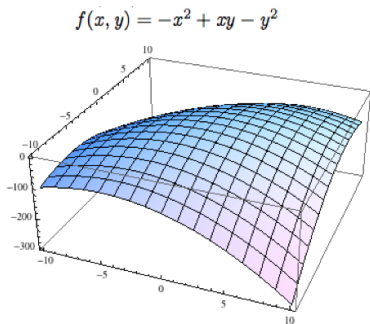
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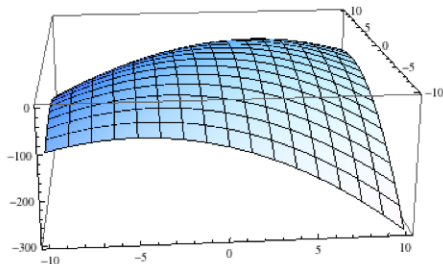
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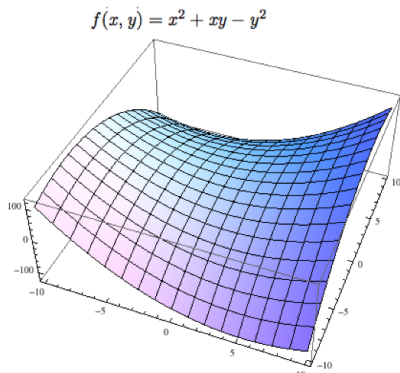
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$$f(x, y) = -x^2 + xy - y^2$$



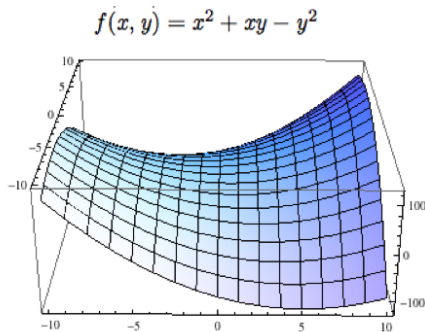
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# Principal Minors

Let  $A$  be an  $n \times n$  matrix. A  $k \times k$  submatrix of  $A$  obtained by deleting any  $n - k$  columns and the same  $n - k$  rows from  $A$  is called a  $k$ th-order *principal submatrix* of  $A$ . The determinant of the  $k \times k$  principal submatrix is called the  $k$ th order *principal minor* of  $A$ .

## Principal Minors: Example

List all the principal minors of the  $3 \times 3$  matrix:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

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- There are also three first order principal minors  $a_{11}$ ,  $a_{22}$ , and  $a_{33}$ .

## Leading Principal Minors

The *leading principal minor* is the determinant of the *leading principal submatrix* obtained by deleting the last  $n - k$  rows and columns of an  $n \times n$  matrix  $A$ . For example list the first, second, and third order leading principal minors of the  $3 \times 3$  matrix:

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- If some  $k$ th order leading principal minor of  $A$  is nonzero but does not fit either of the above sign pattern, then  $A$  is indefinite.

## Testing for Semidefiniteness.

In the previous slide, verifying if  $A$  is positive definite or negative definite depends on strict inequalities. If these inequalities are relaxed we know that the  $A$  is not positive or negative definite. It may still be positive or negative semidefinite. To verify this, we look for the following conditions

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Notice that for determining semi-definiteness, we can no longer check just the leading principal minors, but we must check all principal minors.

# Eigenvalues

Let  $A$  be a square matrix. An *eigenvalue* is a number  $\lambda$  such that

$$(A - \lambda I)\mathbf{x} = 0, \text{ for some } \mathbf{x} \neq 0$$

Also notice that this implies that an eigenvalue is a number  $\lambda$  such that  $A\mathbf{x} = \lambda\mathbf{x}$ .

## Eigenvalues and Singularity

For the system  $(A - \lambda I)\mathbf{x} = 0$ , note that  $\mathbf{x} = 0$  is always a solution. The definition of an eigenvalue requires that there is another solution to  $(A - \lambda I)\mathbf{x} = 0$ . This means that the system  $(A - \lambda I)\mathbf{x} = 0$  has more than one solution. What does this tell us about the matrix  $(A - \lambda I)$ ?

- We know that when the eigenvalue  $\lambda$  is subtracted from the diagonal elements of  $A$ , the resulting matrix is singular. Therefore, we know that  $\det(A - \lambda I) = 0$



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- We know that when the eigenvalue  $\lambda$  is subtracted from the diagonal elements of  $A$ , the resulting matrix is singular. Therefore, we know that  $\det(A - \lambda I) = 0$
- Also we know that  $A$  is non-singular if and only if 0 is not an eigenvalue of  $A$ .

## Eigenvalue: Example

Find the eigenvalues of the matrix

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{pmatrix}$$

To find the eigenvalues we must solve for  $\lambda$  in the following equation

$$\left| \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = 0$$

## Eigenvalue: Example Continued

$$\left| \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = 0$$

$$\Rightarrow \left| \begin{pmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 4 & -17 & 8 - \lambda \end{pmatrix} \right| = 0$$

$$-\lambda^3 + 8\lambda^2 - 17\lambda + 4 = 0$$

$$-(\lambda - 4)(\lambda^2 - 4\lambda + 1) = 0$$

So one eigenvalue is  $\lambda = 4$ .

## Eigenvalue: Example Continued

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So one eigenvalue is  $\lambda = 4$ . The other is obtained by using the quadratic formula

$$\lambda = 4 \pm \frac{\sqrt{(-4)^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1} = 2 \pm \sqrt{3}$$

The eigenvalues are  $\lambda = \{2 - \sqrt{3}, 2 + \sqrt{3}, 4\}$ .

# Eigenvectors

Given an eigenvalue  $\lambda$ , an *eigenvector* associated with  $\lambda$  is a non-zero vector  $\mathbf{x}$  such that  $(A - \lambda I)\mathbf{x} = 0$ . Notice that eigenvectors are only defined up to a scalar: If  $(A - \lambda I)\mathbf{x} = 0$ , then also  $(A - \lambda I)2\mathbf{x} = 0$ . So all multiples of  $\mathbf{x}$  are also eigenvectors associated with  $\lambda$ .

## Intuition of Eigenvector

Eigenvectors are vectors that, when multiplied by  $A$ , end up as scalar multiples of themselves. The transformed vector points in the same direction as the original vector, but may vary in length. Consider the matrix.

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

Then  $A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ x + 3y \end{pmatrix}$

- $A \begin{pmatrix} 1 \\ 1 \end{pmatrix}$  is an eigenvector for the eigenvalue of 4, since

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}.$$

## Intuition Continued

Something like  $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$  is not since  $A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

## Eigenvectors: Example

Find the eigenvectors of the matrix

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

What is the first step?



## Eigenvectors: Example

Find the eigenvectors of the matrix

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What is the first step? First we must find the eigenvalues of the matrix. The determinant of the matrix  $(A - \lambda I)$  is

$$-\lambda(2-\lambda)(3-\lambda)+2(2-\lambda) = -\lambda(6-5\lambda+\lambda^2)+4+2\lambda = -8\lambda+5\lambda^2-\lambda^3+4$$

Thus we must find  $\lambda$  such that the determinant is equal to zero or

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 1)(\lambda - 2)(\lambda - 2) = 0$$

This means that the eigenvalues are  $\{1, 2, 2\}$ .

## Eigenvectors: Example

We have calculated the eigenvalues to be  $\{1, 2, 2\}$ . Note that each eigenvector corresponds to an eigenvalue. First consider the eigenvalue  $\lambda = 1$ . The matrix  $A - \lambda I$  is then

$$\begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$$

This implies that the system of equations corresponding to  $(A - \lambda I)\mathbf{x} = \mathbf{0}$  is

$$x_1 = -2x_3$$

$$x_1 = -x_2 - x_3$$

$$x_1 = -2x_3$$

## Eigenvectors: Example

Using the system of equations we can solve for the eigenvector.

$$x_1 = -2x_3$$

$$x_1 = -x_2 - x_3$$

$$x_1 = -2x_3$$

This implies that  $-2x_3 = x_1 = -x_2 - x_3 \implies x_3 = x_2$  and we have

And so

$$x_1 = -2x_3$$

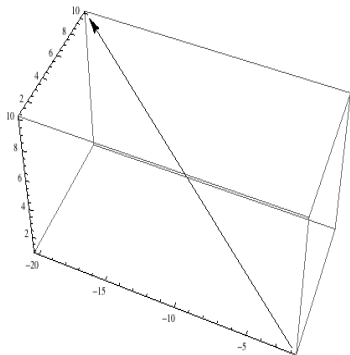
$$x_2 = x_3$$

$$x_3 = x_3$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} x_3$$

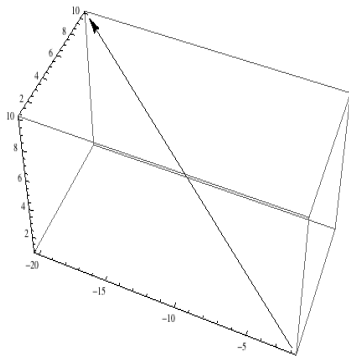
## Eigenvector: Example

The eigenvector is defined up to a scalar. So we can let  $x_3$  be any value we want, say  $x_3 = s$  and trace out the line in 3 dimensional space. The direction of this line is the eigenvector of  $\lambda = 1$ . For each point on this line it is the case  $(A - \lambda I)\mathbf{x} = 0$ .



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This implies

$$x_1 = -x_3$$

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Which implies

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ ? \\ -1 \end{pmatrix} x_1$$



## Eigenvector: Example

Since the system that corresponds to  $\lambda = 2$  is independent of  $x_2$ , we can let both  $x_1$  and  $x_2$  be anything we want. Say  $x_1 = s$  and  $x_2 = t$ . Then we can write the solution to the system as

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} s + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} t$$

The eigenvectors that correspond to  $\lambda = 2$  are  $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$  and

$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$  and their linear combinations.

## Eigenvector: Example

To check that the eigenvectors and their linear combinations correspond to  $\lambda = 2$  we can just verify that for each eigenvector  $(A - \lambda I)\mathbf{x} = 0$ .

$$(A - \lambda I)\mathbf{x} = \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 + 0 + 2 \\ 1 + 0 - 1 \\ 1 + 0 - 1 \end{pmatrix} = 0$$

$$(A - \lambda I)\mathbf{x} = \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 + 0 + 0 \\ 0 + 0 + 0 \\ 0 + 0 + 0 \end{pmatrix} = 0$$

# Steps for finding the Eigenvalue and Eigenvector

Steps for finding the eigenvalues that correspond to the  $n \times n$  matrix  $A$ .

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- Chose a  $\lambda$  and compute  $(A - \lambda I)$
- Identify the system of equations that corresponds to  $(A - \lambda I)\mathbf{x} = 0$
- Solve for all of the elements of the vector  $\mathbf{x}$  so that they are defined in terms of one element of this vector. Use this to define the eigenvectors.

## Number of Eigenvalues and Eigenvectors

Finally note that the equation defining  $\lambda$  is a for a  $n \times n$  matrix is a  $n$  order polynomial equation. From our first example,

$$\left| \begin{pmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 4 & -17 & 8 - \lambda \end{pmatrix} \right| = 0$$

$$-\lambda^3 + 8\lambda^2 - 17\lambda + 4 = 0$$

This also means that there are  $n$  roots and thus  $n$  eigenvalues for a  $n \times n$  matrix, though some may not be real. Each eigenvalue has a corresponding eigenvector. If an eigenvalue is repeated  $k$  times then it will have  $k$  corresponding eigenvectors.

## Diagonalizability

A square matrix  $A$  is called *diagonalizable* if there exists an invertible matrix  $P$  such that  $P^{-1}AP$  is a diagonal matrix. We state without proof the following theorem (For proof see SB appendix of chap 23). Let  $A$  be a  $n \times n$  matrix and assume that  $A$  has  $n$  linearly independent eigenvectors. Define the matrices

$$D = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}, P = \begin{pmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n \end{pmatrix}$$

where  $\mathbf{p}_i$  is the  $i$ th linearly independent eigenvector. Then  $D = P^{-1}AP$  and  $PDP^{-1} = A$ .